

Technical Master Compliance & Safety Report: MCU (Motor Control Unit / VCU)

Parameter	Specification
Vehicle Subsystem	Motor Control Unit (MCU / VCU - Central Vehicle Control Unit)
Production Target	MCU_FW (ESP-IDF v5 + FreeRTOS SMP)
Applied Standards	MISRA-C:2012, ISO 26262 (ASIL-D / ASIL-B Principles), Zero Dynamic Memory, 100 Hz Deterministic FreeRTOS
CI/CD Build & Test Status	PASSED (Compilation with -Werror , static analysis, and 100% host unit tests passing)

1. Executive Summary & System Architecture

The **Motor Control Unit (MCU / VCU)** is the central safety-critical supervisor and high-level vehicle controller of the Formula Student electric racing car:

- Redundant Dual Accelerator Pedal Position Sensing (APPS 1 & APPS 2):** Median-of-3 filtering, wire-break and short-circuit boundary checking ($\pm 15\%$), continuous sensor plausibility checking ($< 10\%$ discrepancy) with a strict 100 ms grace window in accordance with Formula Student Electric vehicle regulations.
- Hydraulic Brake Pressure (HPS) & Dual-Segment Steering Sensing:** High-speed analog acquisition with calibrated deadbands and threshold triggers.
- Brake System Plausibility Device (BSPD) Interlock:** Hard torque cutoff to 0.0 Nm whenever accelerator demand exceeds 25% while the mechanical brakes are actuated (> 100 ADC counts), latching until pedal demand drops below 5%.
- Ready-To-Drive (R2D) State Machine:** Secure multi-step arming sequence with Ready-To-Drive Sound (RTDS) acoustic horn timer (2000 ms).
- IRAM Hardware Interrupt Wheel Speed Processing:** Direct microsecond pulse period calculation across 4 phonic wheel Hall sensors (ENC_FL, ENC_FR, ENC_RL, ENC_RR) with 150 ms zero-speed timeout.
- Advanced Torque Management & Drive Modes:**
 - Mode 1 (ECO / 40 kW)** and **Mode 2 (AutoX / 72 kW)** torque maps.
 - Mode 3 (Endurance / 22 km):** Closed-loop slow PI energy-budget controller with power slew rate limiter (3 kW/s), back-calculation anti-windup, and RPM-dependent power limits.
 - Mode 8 (Launch Control):** Target slip ratio regulation in static and dynamic phases.
 - Regenerative Braking:** Fixed and brake-pressure proportional modes.
 - Predictive Battery Sag Protection:** Adaptive internal resistance (R_{int}) and Open-Circuit Voltage (OCV) estimator preventing BMS undervoltage shutdowns.
- Dual-Bus CAN Architecture:**
 - Inverter TWAI (500 kbps):** High-speed 100 Hz cyclic torque commands (REG_TORQUE = 0x90, 0 \dots 32767) to Unitek Bamocar.
 - Car CAN (500 kbps):** Real-time telemetry broadcasting (0x020, 0x021, 0x200) and DTC diagnostics on 0x502.

2. Exhaustive Fault Management Matrix (DTC & Safe States)

The MCU firmware implements a dedicated fault manager (fault_manager.c) providing subsystem isolation, safe state latching, and CAN DTC broadcasting:

DTC Code	Fault Identifier	Category	Priority	Exact Trigger Condition	System & Physical Hardware Reaction	Recovery / Reset Mechanism	Broadcaster Frame
101	FAULT_CODE_APPS_IMPLAUSIBLE	FAULT_CAT_HARDWARE	HIGH	Discrepancy between APPS 1 and APPS 2 exceeds 10% for > 100 ms (100,000\ μ s grace period, FS Rules).	Immediately clamps torque command to 0.0 Nm , disables inverter drive command, and latches APPS subsystem in fault_manager_lock_subsystem(FAULT_SUBSYS_APPS) .	Automatic recovery when the discrepancy between both sensor signals returns below 10%.	CAN ID 0x502 0 = 1 (Hig Active) 1 = 0 (HW returns below 10% Byte 1 (High Pri Byte = 101 7 = Subsys: Bitmask (E APPS)
102	FAULT_CODE_APPS_WIRE_BREAK	FAULT_CAT_HARDWARE	HIGH	Analog ADC voltage on APPS 1 or APPS 2 exceeds $\pm 15\%$ safety margins outside calibrated travel limits (indicates short to 3.3V or severed ground wire).	Immediately clamps torque to 0.0 Nm and prohibits drive actuation.	Requires analog voltage to return within calibrated physical limits.	CAN ID 0x502 0 = 1 By 4 = 102
103	FAULT_CODE_BRAKE_SENSOR_ERR / DISCONNECT	FAULT_CAT_HARDWARE	HIGH	Hydraulic pressure sensor (HPS) disconnected or raw ADC outside valid electrical bounds (< 50 or > 4000 counts).	Latches brake subsystem mask (FAULT_SUBSYS_BRAKES), blocks entry into R2D driving state, and forces safe torque limiting.	Automatic recovery once valid electrical signal is re-established.	CAN ID 0x502 0 = 1 By 4 = 103

DTC Code	Fault Identifier	Category	Priority	Exact Trigger Condition	System & Physical Hardware Reaction	Recovery / Reset Mechanism	Broadcaster Frame
104	FAULT_CODE_TWAI_BUS_OFF	FAULT_CAT_COMMUNICATION	HIGH	Inverter CAN or Car CAN bus enters BUS_OFF state.	Initiates automatic hardware recovery via twai_initiate_recovery() , commands 0.0 Nm torque, and resets internal mailboxes.	Automatic once physical bus resynchronization is achieved.	CAN ID 0x! immediately bus recover
105	FAULT_CODE_BMS_SAG_LIMIT	FAULT_CAT_RESOURCES	LOW	High-voltage battery pack voltage drops near minimum discharge threshold during aggressive acceleration.	Adaptive torque derating based on estimated internal resistance R _i (int) and OCV to prevent tripping accumulator BMS shutdown.	Dynamic recovery as battery open-circuit voltage stabilizes.	CAN ID 0x502 0 = 2 (Wa Byte = 105
106	FAULT_CODE_BSPD_TRIPPED	FAULT_CAT_HARDWARE	HIGH	Accelerator pedal > 25% while hydraulic brake is actuated (> 100 ADC counts).	Hard clamp of torque demand to 0.0 Nm (mandatory Formula Student EV rule EV4.7).	Latched: Does not clear until accelerator pedal is completely released to < 5%.	CAN ID 0x502 0 = 1 Byte 4 = 106

3. CAN / TWAI Bus Network Architecture (500 kbps)

3.1. Transmitted CAN Frames

CAN ID	Message Name	DLC	Rate	Target Nodes	Payload Layout (Byte-by-Byte)	Resolution & Units
0x0C0	INVERTER_TORQUE_CMD	8	100 Hz	Unitek Bamocar Motor Inverter	Byte 0 : REG_TORQUE (0x90) Byte 1-2 : Torque Demand (Little-Endian int16) Byte 3..7 : Inverter Command Control	32767 = 100% nominal motor torque
0x020	MCU_WHEEL_SPEEDS	8	100 Hz	Dashboard, Telemetry, Data Logger	Byte 0-1 : RPM Front-Left (BE uint16) Byte 2-3 : RPM Front-Right (BE uint16) Byte 4-5 : RPM Rear-Left (BE uint16) Byte 6-7 : RPM Rear-Right (BE uint16)	1\ LSB = 1\ RPM Range: 0 \dots 10000\ RPM
0x021	MCU_VEHICLE_STATE	8	100 Hz	Dashboard, ECU, PDM, Telemetry	Byte 0-1 : Steering Angle (BE int16, 0.1^°/LSB) Byte 2-3 : Front Brake Pressure (BE uint16) Byte 4-5 : Rear Brake Pressure (BE uint16) Byte 6 : Brake Switch & R2D State (4 = R2D) Byte 7 : Demanded Torque Percentage (0 \dots 100\%)	Byte 6: Bit 0 = Brake Active, Bits 1-3 = R2D State
0x200	MCU_LOG_TELEMETRY	8	10 Hz	Data Logger, Live Telemetry	Internal controller states and diagnostic logging	Structured telemetry payload
0x502	MCU_DIAGNOSTIC_DTC	8	10 Hz / On-Fault	Safety Master, Dashboard, Telemetry	Byte 0 : High Fault Active (1 = Yes, 0 = No) Byte 1 : Category Byte 2 : Priority Byte 3-4 : Active DTC Code (Big-Endian uint16) Byte 5-6 : Fault Count Byte 7 : Locked Subsystems Bitmask	Bit 0: APPS, Bit 1: Brakes, Bit 2: Inverter, Bit 3: CAN Car

3.2. Received CAN Frames

CAN ID	Sender Node	Key Payload Fields	MCU Action & Functional Behavior
0x401	ECU (FANS_DYN10)	Motor & Inverter Temperatures (^°C).	Real-time thermal protection: triggers predictive torque derating if motor or inverter exceed critical operating temperature limits.
0x006	PDM	Auxiliary Battery Voltage & Current (12.6V).	Low-voltage system health monitoring.
0x180	Inverter Bamocar	Actual Motor Speed (RPM), Internal Torque, DC Bus Voltage.	Real-time feedback for traction control, launch control slip calculation, and regenerative power limits.

4. Unit Test Validation Suite (Unity Test Framework)

The MCU firmware algorithms are thoroughly validated in native host simulation:

```
Processing * in native environment
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Building...
Testing...
test/test_main.cpp:95: test_apps_calibration_and_deadband [PASSED]
test/test_main.cpp:96: test_bspd_interlock [PASSED]
test/test_main.cpp:97: test_r2d_state_machine [PASSED]
test/test_main.cpp:98: test_torque_modes_and_limits [PASSED]
test/test_main.cpp:99: test_fault_manager_apps_implausibility [PASSED]
----- 5 Tests 0 Failures 0 Ignored -----
OK
```

Detailed Test Case Specifications

- test_apps_calibration_and_deadband :
 - Asserts accelerator mapping across deadband (14.0%), normal travel, and full-throttle points.
- test_bspd_interlock :
 - Simulates brake pedal engagement during acceleration (> 25% APPS): confirms instantaneous torque cutoff to 0.0 Nm and verifies that torque cannot be restored until accelerator is fully released to < 5%.
- test_r2d_state_machine :

- Tests transition sequence: `OFF` \to `WAITING_BRAKE` \to `WAITING_BUTTON` \to `SOUNDING_RTDS` (verifies 2000 ms acoustic timer) \to `READY_TO_DRIVE` .
 - 4. `test_torque_modes_and_limits` :
 - Asserts maximum power and torque clipping for Mode 1 (ECO 40 kW), Mode 2 (AutoX 72 kW), and Launch Control slip regulation.
 - 5. `test_fault_manager_apps_implausibility` :
 - Simulates sensor discrepancy > 10\% for > 100 ms: validates generation of DTC `101` , activation of `FAULT_SUBSYS_APPS` lock, and complete torque nullification.
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5. Industrial Standards & Safety Compliance Certification

1. Zero Dynamic Memory Allocation:

- 100% static allocation. Zero use of dynamic heap (`malloc` , `free` , `new`).
- Static FreeRTOS queues, semaphores, and tasks pinned to dedicated CPU cores.

2. Deterministic Hard Real-Time Timing:

- Core control loop executes at strict 100 Hz (10.0 ms) via `vTaskDelayUntil` .

3. MISRA-C:2012 Strict Conformance:

- Fully qualified integer types, bounded loop counters, explicit casts, and defensive programming throughout.

4. Clean Compiler Verification:

- Zero warnings under `-Wall -Wextra -Werror` compiler flags.